

HIGH-SCALABLE AND SECURE BLOCKCHAIN: GENERATIVE ARTIFICIAL INTELLIGENCE AS A KEY ENABLER

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ABSTRACT

Blockchain has gained increasing attention as a promising technology for reliable communication and secure collaboration in various fields. However, deploying blockchain in communication systems remains challenging due to its limited performance and large costs. To address this challenge, we propose blockchain generative intelligence (BGI) to enhance both scalability and security of blockchain-based communication systems. Firstly, we introduce the three-level optimization for blockchain enhancement. Based on this, we introduce how to leverage Generative AI (GenAI) to address the scalability and security issues within the three levels of blockchain optimization. Furthermore, in the BGI framework, we incorporate existing blockchain scaling strategies and utilize the advantages of GenAI to mitigate the challenges faced by these scaling strategies, including directed acyclic graph (DAG), sharding, protocol optimization and cross-chain technology. Moreover, we conduct a case study demonstrating the effectiveness of deploying BGI-assisted sharding blockchain in autonomous vehicle platooning (AVP), as one of the promising application scenarios of blockchain. Finally, we evaluate the performance of the proposed strategies by simulations.

I. INTRODUCTION

Next-generation communication systems are designed to support billions of interconnected devices, enabling transformative applications such as smart cities, industrial automation, and remote healthcare ecosystems. With the continuous development of artificial intelligence (AI) and Internet of things (IoT), the demand for highly reliable, secure, and efficient communication networks has experienced exponential growth. The proliferation of distributed devices and real-time data exchange presents critical challenges in ensuring data integrity, trustless collaboration, and tamper-proof transactional records. Blockchain, as a revolutionary distributed ledger technology with inherent

tamper-resistance, offers a paradigm shift for establishing decentralized, trust-based frameworks for next-generation communication systems.

Despite remarkable progress in finance, supply chain management and digital identity verification, the application of blockchain in communication systems remains highly challenging. Taking the Internet of vehicles (IoV) as an example, it offers services in diverse scenarios through extensive vehicle to everything (V2X) connections. The primary goal of blockchain is to quickly, safely, and accurately share critical information based on vehicular ad-hoc networks to improve driving safety. However, the scalability of blockchain still needs much improvement, including limited throughput, significant consumption of computational resources, storage, and interoperability issues. In addition, due to decentralization and other features, blockchain itself is also faced with numerous security risks, such as smart contract vulnerabilities, consensus mechanism flaws and 51% attack. Therefore, a high-scalable and secure blockchain is essential for its application in communication systems.

Currently, scalability and security optimization strategies for blockchain systems can be classified into three levels. In the first level, underlying technologies can be optimized, e.g., optimizing computing and communication systems supporting the blockchain. In the second level, the blockchain itself can be improved, e.g., sharding and directed acyclic graph. In the third level, blockchain on-chain applications can be optimized, for example, cross-chain technology [1]. However, traditional optimization strategies face numerous challenges and cannot simultaneously satisfy the requirements for both scalability and security in next-generation communication systems. For example, blockchain sharding technology can effectively improve scalability, but by dividing the network into multiple smaller shards, it makes each individual shard more susceptible to 51% attacks. This highlights the need for a novel approach that can optimize both blockchain scalability and security.

To solve this challenge, we propose the Blockchain Generative Intelligence (BGI) framework, which

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systematically applies Generative Artificial Intelligence (GenAI) to blockchain optimization. Unlike previous flat and non-systematic applications of GenAI in blockchain, the BGI framework organizes GenAI's role into three distinct levels: 1) Level 0: Optimization of the off-chain network. 2) Level 1: Optimization of the blockchain itself. 3) Level 2: Optimization of on-chain applications. Additionally, each level includes two key performance dimensions: scalability and security. This systematic approach provides a clearer blueprint for combining GenAI with blockchain optimization.

This paper aims to conduct a forward-looking exploration of high-scalable and secure blockchain based on BGI for next-generation communication systems, taking autonomous vehicle platooning (AVP) as a use case.

The main contributions of this work are summarized as follows:

- We propose BGI for high-scalable blockchain with the consideration of security. This framework divides blockchain optimization into three levels and two planes, with the aim of systematically applying GenAI to blockchain scalability and security optimization. The goal is to improve blockchain systems in next-generation communication systems and provide a clear blueprint for the combination of GenAI and blockchain optimization.
- In addition to directly using GenAI to enhance blockchain scalability and security, our BGI framework also incorporates representative blockchain scaling strategies, including Directed Acyclic Graph (DAG), sharding, protocol and parameter optimization and cross-chain technology. By leveraging GenAI, we aim to improve the effectiveness of these strategies, enhancing both their performance and security. This offers a novel path for blockchain optimization.
- We present a sharding blockchain-enabled AVP scenario as a case study. Within this use case, we utilize Generative Diffusion Models (GDMs) to optimize the sharding blockchain consensus, thereby validating the effectiveness of the BGI based blockchain optimization.

II. PRELIMINARIES

A. BLOCKCHAIN

Blockchain, as a decentralized and tamper-proof distributed ledger technology, can facilitate secure data sharing and trusted multiparty consensus. Typically, a blockchain system generally comprises six layers: the data layer handles the storage and management of data, including the generation and validation of blocks. The network layer manages inter-node communication. The consensus layer ensures consistency and correctness across all nodes through consensus mechanisms, maintaining system unity. The incentive layer encourages nodes to participate in the maintenance and development of the network through economic incentives. The contract layer supports smart contract execution for automated service logic. Finally, the application layer offers various blockchain-based applications and services for various needs [2].

In optimizing the six layers, blockchain scalability confronts an impossible trinity: the inherent difficulty in simultaneously considering decentralization, security, and scalability. In detail, first, greater decentralization,

achieved by increasing nodes, will increase the complexity and cost of the system. Secondly, enhanced security demands more computational resources and time for transaction validation, which affects the processing speed of the system. Conversely, improving system processing efficiency frequently means sacrificing some degree of decentralization or security. For example, reducing the number of consensus nodes can accelerate transaction confirmation but simultaneously lowers the degree of decentralization, making the system more vulnerable to attacks such as the 51% attack.

B. BLOCKCHAIN SCALABILITY

Scalability is a system's capacity to handle increasing workloads. For blockchain, it profoundly impacts system robustness, performance, and adaptability [3]. Based on the six-layer architecture, blockchain scaling strategies fall into three levels:

Level 0: Level 0 optimization involves improving the underlying network without modifying the blockchain itself, through enhancements like optimized communication protocols, increased network transmission speeds, and refined network topology. For example, in vehicular blockchain networks, dedicated communication protocols such as IEEE 802.11p can be integrated to improve data exchange between vehicles. While broad, its scope often operates independently of the blockchain's core architecture.

Level 1: Level 1 optimization focuses on enhancing core blockchain technology by adjusting its data, network, consensus, and incentive layers. Common scalability solutions such as DAG and sharding fall within the scope of Level 1, e.g., Ethereum 2.0's sharding, which alters the mechanism of the network layer to divide the blockchain into smaller, parallel chains for improved transaction throughput.

Level 2: Level 2 optimization refines upper-layer applications, specifically the contract and application layers. This encompasses designing and deploying more efficient smart contracts or employing task off-loading strategies to boost blockchain performance. A prime example is Ethereum's adoption of rollups, where transactions are processed off-chain and then submitted in batches to the main chain, significantly reducing congestion and gas fees.

C. BLOCKCHAIN SECURITY

In addition to scalability, security is another crucial metric for a blockchain. Within our BGI framework, we also divide blockchain security into three levels.

Level 0: This level primarily addresses the security of the blockchain's off-chain network. These networks often face security issues similar to those of traditional networks, such as encryption and network attacks. Therefore, traditional network security measures are also applicable to level 0 optimization. However, traditional methods rely on predefined rules for known vulnerabilities or attack patterns, leaving the off-chain network vulnerable to "unknown" or "zero-day" attacks that are difficult for conventional security tools to detect or address.

Level 1: The scope of level 1 security includes transaction data security, consensus security, and the security of the blockchain itself. Despite its inherent advantages, such as the tamper-proof and traceable nature of on-chain data, blockchain technology also faces transaction and consensus security issues such as fraudulent transactions, Sybil attacks, and 51%

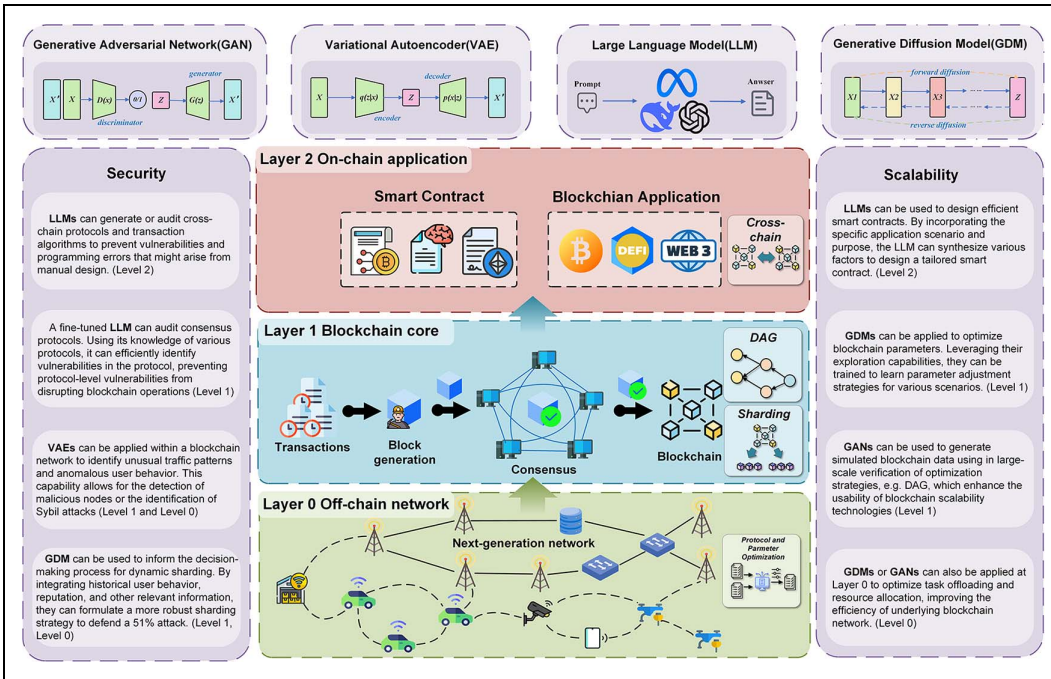


FIG. 1. Schematic diagram of BGI. The BGI categorizes blockchain optimization into three levels: off-chain network, blockchain core, and on-chain applications. The off-chain network supports the deployment and operation of the blockchain, while the blockchain itself underpins diverse upper-layer applications, such as decentralized finance (DeFi) and Web3. Within each of these levels, GenAI plays a unique and crucial role in optimizing two key performance dimensions: scalability and security.

attacks due to its decentralized nature and consensus process.

Level 2: This level focuses on the security of on-chain applications and smart contracts. Security vulnerabilities or logical errors in smart contracts, in particular, are often difficult to detect but can severely impact the blockchain's upper-layer applications, making them a primary security concern in level 2. Traditional security methods, such as manual code audits and static analysis tools that rely on predefined rules, are often slow, costly, and struggle to keep up with the rapidly evolving threat landscape.

The purpose of the BGI framework is to integrate GenAI into the three levels of blockchain optimization. Using the unique advantages of GenAI, this framework provides a novel approach for scalability optimization while addressing the specific security challenges that exist in each of these three levels, going beyond existing methods.

III. OVERVIEW OF BLOCKCHAIN GENERATIVE INTELLIGENCE FRAMEWORK

As shown in Fig. 1, the BGI framework divides the scalability and security optimization of blockchain in next-generation communication networks into three levels. In this section, we first introduce the four primary GenAI models within the BGI framework. Following that, we will systematically describe the roles that GenAI plays within the three levels of blockchain optimization.

A. GENAI MODELS IN BGI

1) **Large Language Model (LLM):** An LLM is trained on vast amounts of text data based on deep learning architectures, aiming to understand and generate human language. Currently, an LLM has achieved remarkable performance in various tasks

such as text generation, planning, and coding. In BGI, LLMs can be applied in smart contract auditing and cross-chain protocol design.

2) **Generative Adversarial Network (GAN):** A GAN is composed of a generator $G(z)$ that generates realistic samples from a latent space, and a discriminator $D(x)$ that learns to differentiate between real and generated data. This adversarial process refines the generator's ability to create high-quality synthetic data. Within BGI, GANs are vital for data augmentation and compression. For instance, the feature extraction and analysis prowess of GAN discriminators can be used to compress data stored in blockchain nodes, thus enhancing the efficiency of data storage and transmission [4].

3) **Variational Autoencoder (VAE):** VAE is a probabilistic generative model that learns the latent distribution of data to generate new samples. Unlike GAN, VAE maximizes data likelihood by encoding x into a lower-dimensional latent space z , then reconstructs the data via a decoder. This compression enables VAE to generate continuous, diverse samples. VAE can detect unknown blockchain anomalies. For example, [5] combines VAE with LLM to model normal network states, identifying anomalies and significantly improving advanced persistent threat detection, crucial for large-scale blockchain scalability and security.

4) **Generative Diffusion Model (GDM):** GDM is a type of probabilistic generative model composed of forward and reverse diffusion processes in the Markov chain. Due to their unique iterative denoising process, GDM has strong exploration and capture capabilities compared to GAN and VAE [6]. Therefore, in BGI, GDM can be used for dynamic optimization and adjustment of protocols and parameters. In Section V, we present a use case of using GDM to optimize blockchain consensus.

B. GENAI-BASED BLOCKCHAIN OPTIMIZATION IN BGI

The BGI framework divides blockchain optimization into three levels: the off-chain network (level 0), the blockchain itself (level 1), and on-chain applications (level 2). Optimization for each level is further categorized into two planes: security and scalability. The core principle of BGI is to systematically apply GenAI throughout these levels to enhance the entire system's scalability and security.

1) **Level 0 Optimization:** Level 0 optimization refers to enhancing the off-chain network, which includes environments such as the IoV, the Internet, and the IoT. By improving the scalability and security of the off-chain network, it can better support the operation of higher-level blockchains and the deployment of blockchain applications. GenAI also has a unique role to play in level 0.

Scalability: The communication efficiency of a blockchain's off-chain network directly impacts the transaction speed and consensus efficiency of its higher layers. In the off-chain network, GenAI, such as GDMs, can be applied to optimize wireless network resource allocation and task offloading. Using their superior exploration and generative capabilities, GDMs can learn the dynamic changes of the state of the wireless network and formulate strategies for resource allocation or task offloading based on the current state, thus improving network resource utilization and overall network efficiency [7].

Security: Regarding off-chain network security, GenAI, such as GANs, can be used for network anomaly detection. Unlike traditional methods that can only detect predefined attack patterns, GenAI learns normal traffic and transaction patterns to build a secure network model. This enables the detection of "unknown" or "zero-day" attacks. Simultaneously, a GAN generator can create "adversarial inputs" that mimic normal behavior but contain subtle malicious patterns. These inputs can be used to test network protocols or train security models. This method provides a powerful, automated "red teaming" tool that proactively generates attack patterns to continuously evaluate and strengthen the defensive capabilities of a blockchain network.

2) **Level 1 Optimization:** Level 1 blockchain optimization refers to enhancing the core components of blockchain, including the transaction process, block generation, consensus mechanism, and the blockchain's inherent structure. Optimizing this level directly leads to significant improvements in blockchain scalability and security, which is why traditional blockchain scaling strategies have historically focused on this level. However, GenAI can further enhance blockchain performance compared to traditional scaling strategies.

Scalability: When a blockchain is applied to scenarios like IoV or IoT, limited storage and network resources often become a bottleneck for scalability. GenAI, specifically VAEs, can utilize their encoding and compression capabilities to perform efficient lossy compression on on-chain data. By compressing complex transaction histories or account data into low-dimensional latent vectors, they can significantly reduce both storage space consumption and synchronization time, improving the applicability of blockchain in resource-constrained scenarios such as edge computing. Furthermore, GDMs, which are used for decision-making at Level 0, can also be applied to dynamically adjust the blockchain's own

parameters, such as block size and block generation intervals. This enhances the blockchain's scalability across various heterogeneous application scenarios.

Security: Auditing transaction security is a key aspect of ensuring a blockchain's secure operation. VAEs can be used to achieve highly efficient transaction auditing. A VAE operates by learning the probability distribution of normal transaction data within a compressed latent space. When a new transaction is introduced, if it deviates significantly from the learned "normal" pattern, the decoder will be unable to accurately reconstruct it, resulting in a high "reconstruction error". This error then serves as a reliable indicator of an anomaly. For example, the Anomaly VAE-Transformer model uses a VAE to extract short-term local information, while a Transformer identifies long-term data dependencies, allowing real-time detection of anomalous behavior in decentralized finance (DeFi) protocols [8].

3) **Level 2 Optimization:** Level 2 blockchain optimization primarily focuses on enhancing the security and efficiency of upper-layer applications, including smart contracts and other blockchain applications.

Scalability: Smart contracts are essential for implementing business logic on a blockchain, but their development faces numerous challenges: complex specifications, high risk of major security vulnerabilities from minor errors, and long development cycles. GenAI, specifically LLMs, can address these issues by using their natural language understanding and code generation capabilities. They can transform human-readable descriptions of business logic into efficient, secure, and scalable smart contract code templates, thereby improving development efficiency. Furthermore, LLMs can analyze existing contracts to identify redundant and computationally intensive sections. By refactoring these parts, they can reduce unnecessary reads and writes to the storage or optimize computational logic, reducing execution costs and improving transaction processing efficiency [9].

Security: LLMs also play a crucial role in level 2 security optimization. In addition to designing efficient and secure smart contracts, LLMs can be used for smart contract auditing. By training on a vast corpus of smart contract code, LLMs can detect vulnerabilities. Unlike simple pattern recognition, they are capable of "line-level reasoning" and can understand the "causal pathways" behind a contract's business logic. For example, the SmartLLM framework [10] leverages Retrieval-Augmented Generation (RAG) and a fine-tuned LLaMA 3.1 model to achieve a 70% accuracy and a 100% recall rate in smart contract vulnerability detection. This performance significantly exceeds that of traditional static analysis tools such as Mythril and Slither.

Furthermore, LLMs can be applied to on-chain applications for transaction analysis and fraud prevention. By analyzing on-chain transaction graphs and user behavior patterns, LLMs can efficiently detect high-risk transactions. This capability is crucial for addressing the analytical challenges posed by the pseudo-anonymity and massive data volume inherent in public blockchains [11].

IV. BGI FOR SCALABLE AND SECURE BLOCKCHAIN

In addition to directly utilizing GenAI to optimize the blockchain, the BGI framework also incorporates mainstream blockchain scaling strategies. This section discusses how GenAI can be applied to the

Key Technologies	DAG (Directed Acyclic Graph)	Protocol and Parameter Optimization	Sharding	Cross-Chain Technology
Definition	DAG allows transactions to exist in the form of a graph. Each transaction can directly or indirectly point to the previous transactions, and there are no circular paths in the graph.	It optimizes the consensus protocol, network protocol, smart contract, etc. of the blockchain, and adjusts blockchain parameters such as block size and block generation intervals.	Sharding divides the blockchain network into multiple smaller sub-networks, each of which is a shard.	Cross-chain refers to realizing the interconnection and intercommunication between different blockchains, breaking the "island" effect between blockchains.
Based on Which Level of Scaling	Level 1 optimization.	Level 1 optimization, and partly involves Level 0 optimization.	Level 1 optimization.	Level 2 optimization.
Targeted Performance Indicators	Throughput	Throughput, cost and consensus security	Throughput and storage	Interoperability, Cross-chain transaction security and privacy
Advantages	Transactions can be processed in parallel, with a high degree of decentralization.	High flexibility, good compatibility, and low cost.	Reduces the burden on nodes and improves transaction speed.	Expands the application scenarios, improves resource utilization, and enhances the interoperability of the system.
Disadvantages	High security verification complexity, difficult consensus mechanism design.	Limited optimization effect, requires professional knowledge, and is difficult to balance performance and security.	Complex cross-shard communication, weak decentralization.	High technical difficulty, cross-chain transaction security risks, and complex supervision.
The Role of BGI	Security verification optimization, consensus algorithm optimization, and simulated data generation.	Protocol design optimization, parameter automatic adjustment.	Shard distribution optimization, cross-shard communication coordination, and security protection.	Cross-chain protocol design, security protection, and cross-chain data analysis.
Typical Cases	Nano ¹ , Byteball ² , and IOTA ³	Ethereum's Gas mechanism optimization and SeaWit	Ethereum 2.0 ⁴ , Zilliqa ⁵	Polkadot ⁶ , Cosmos ⁷
Application Scenarios	Fast transaction settlement, such as charging fee settlement and shared payment, etc. Real-time data sharing, fast and parallel verification and storage.	Customized for different scenarios, safeguarding the security and stability of blockchain-based communication systems.	Large-scale data node management and parallel processing of traffic data.	Multi-chain integration: deploying different blockchains to manage different businesses, achieving cross-platform data sharing.

TABLE I. Comparison of key scaling strategies in BGI.

mainstream blockchain scaling strategies to address the associated scalability and security challenges. Table I presents a summary of the advantages and disadvantages of four main blockchain scaling strategies.

DAG: Given the difficulty in long-term validation of DAG-based ledger security, GenAI, such as GAN, can be used to generate synthetic data that closely resembles real transaction scenarios for simulation testing of the DAG-based ledger system [12]. This allows for the evaluation of performance metrics, such as throughput and security before system deployment, thereby improving the development and testing efficiency of DAG-based ledgers. Moreover, GenAI can be utilized in the management of DAG. For example, based on the real-time status of the network, such as the activity level of nodes and network bandwidth, GenAI can dynamically adjust the parameters of the consensus algorithm or generate new consensus strategies [6]. This improves the security and efficiency of consensus and mitigates the risks of malicious attacks in DAG-based blockchain.

Protocol and Parameter Optimization: In response to trade-off issues in blockchain protocol optimization, GenAI, such as GDMs, can leverage its exploration and generation capabilities to provide dynamic adjustment solutions based on an in-depth understanding of the blockchain parameters and extensive experimental data. For example, a

GDM-based PBFT consensus optimization method is proposed in [12]. By training on blockchain data, the GDM can associate specific conditions with the optimal solutions, then dynamically generating appropriate PBFT configurations, e.g., block size and block generation interval, to maximize blockchain performance in a given scenario.

Sharding: By leveraging multi-dimensional data such as performance, resource status, and geographical location of nodes, GenAI, such as GAN, with the ability of capturing complex distributions can be utilized for tasks like shard task offloading and resource allocation [13]. This ensures load balancing among shards and prevents performance degradation caused by overloaded shards. Secondly, by comprehensively considering factors such as blockchain node reputation scores, account historical behavior, and other relevant information, GDMs can formulate a robust sharding strategy. This prevents an excessive number of potentially malicious nodes from appearing in the same shard, thus mitigating both Sybil attacks and 51% attacks.

Cross-Chain Technology: LLM can assist in designing more efficient and secure cross-chain protocols. By analyzing the characteristics and requirements of different blockchains, it can generate new cross-chain communication and interaction protocols to achieve efficient interoperability between different

¹ <https://nano.org/en>² <https://obyte.org/>³ <https://obyte.org/>⁴ <https://ethereum.org/en/>⁵ <https://www.zilliqa.com/>⁶ <https://polkadot.com/>⁷ <https://cosmos.network/>

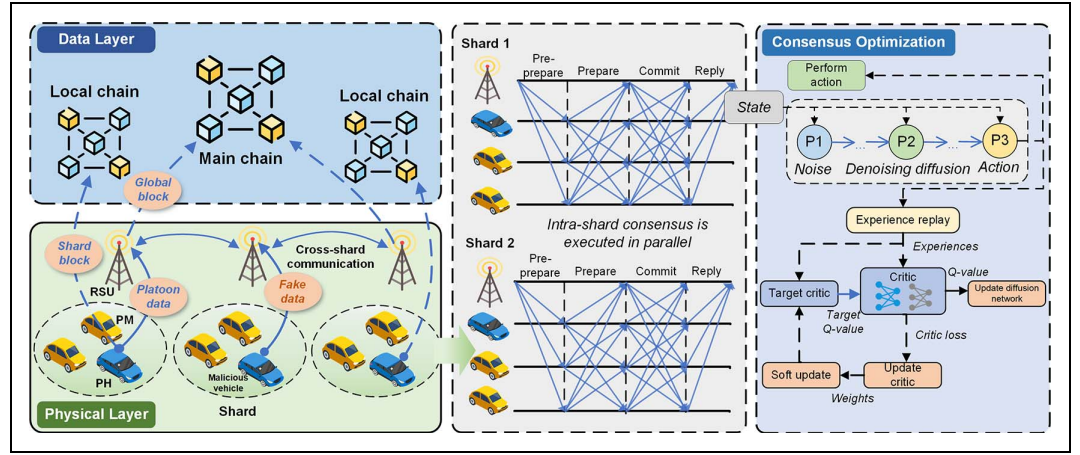


FIG. 2. GDM-based sharding blockchain consensus optimization framework for AVP. The platooning decisions generated by PH must undergo a two-stage consensus process: intra-shard consensus and global consensus, before being ultimately recorded on the main chain. Subsequently, the agent generates a decision to adjust the consensus parameters for the next round based on the environmental state S . After executing this action, the agent receives an environmental reward R . The agent's update method is illustrated in the accompanying figure.

chains and avoid vulnerabilities of manual design. Furthermore, GenAI like VAEs can be used to mine and analyze cross-chain data. By leveraging their ability to learn compressed representations, VAEs can efficiently analyze vast amounts of data in cross-chain transactions, extracting key information, or generating summary reports. This capability supports scenarios such as market research and transaction auditing, thereby promoting the development of DeFi [14].

V. CASE STUDY: BLOCKCHAIN OPTIMIZATION FOR AUTONOMOUS VEHICLE PLATOONING

Autonomous vehicle platooning (AVP) is a transportation mode where vehicles travel in a coordinated formation, which can reduce the complexity of traffic management and minimizing energy consumption. In AVP, blockchain is considered an effective technology for data management and sharing. However, the scalability and security issues of blockchain increase the difficulty of its application in AVP.

In this section, we propose a sharding blockchain-based data-sharing architecture for AVP, using GDM to enhance the throughput performance of the sharding blockchain system while ensuring low transaction latency.

With this architecture, different shards can perform local consensus in parallel, significantly boosting overall blockchain throughput and reducing storage resource consumption. As shown in Fig. 2, the framework consists of two primary layers:

Physical Layer: The physical layer includes vehicles and roadside units (RSUs). The AVP is divided into M shards, and each shard comprises a single vehicle platoon. Vehicles in a platoon are mainly categorized into two types: platoon head (PH) and platoon member (PM). The PH is responsible for formulating platooning decisions, such as the platoon's speed and inter-vehicle spacing. Let the set of PHs be denoted as $\mathcal{H} = \{h_1, h_2, \dots, h_M\}$.

Data Layer: Its primary function is platoon data storage and management. Two types of chain are present within the sharding blockchain: local chains and the main chain. The local chain is deployed within each shard. The main chain, consisting of the RSUs

from different shards, is responsible for managing the information of the local chains.

Consensus Process: When a PH formulates a platoon decision, it packages the decision into a shard block. This block then undergoes consensus within the platoon to be recorded on the local blockchain. Afterward, the PH submits the shard block to the RSU. The block-generation RSU then packages the shard blocks from each shard into a larger block. This global block is recorded on the main chain after undergoing global consensus among the RSUs.

Attack Model: Several Malicious Vehicles (MV) are in the AVP system. Cross-shard transactions require consensus among all PHs. According to the PBFT protocol, if the number of MVs in $\mathcal{H}$ exceeds $(M-1)/3$, these malicious vehicles could potentially launch a 51% attack, leading to consensus failure.

A. PROBLEM FORMULATION

In AVP, several parameters influence blockchain performance: vehicle computing resources, communication rate, and average transaction size. Vehicle computing resources, e.g., CPU speed, affect how quickly vehicles can package blocks. The communication rate affects the propagation speed of the consensus messages between nodes. The average transaction size affects the adjustment of the block size and block generation interval. Together, these factors jointly determine the throughput and latency of the blockchain consensus process.

Optimization Concept: The goal of optimizing the consensus process is to improve blockchain throughput while ensuring cross-chain transaction security. To achieve this, PH $\mathcal{H}$, block size S_B , and block generation interval I_B must be adjusted based on vehicle computing resources C , communication rate R , and transaction size S_t , where the choice of $\mathcal{H}$ affects the block generation speed and the consensus success rate. In other words, the task is to find the optimal parameters $\{\mathcal{H}, S_B, I_B\}$.

B. CONSENSUS OPTIMIZATION METHOD

Fig. 2 illustrates our GDM-based consensus optimization algorithm. The model workflow is as follows:

1) **Environmental State Modelling:** We first model the AVP environment, extracting parameters such as vehicle computing resources and

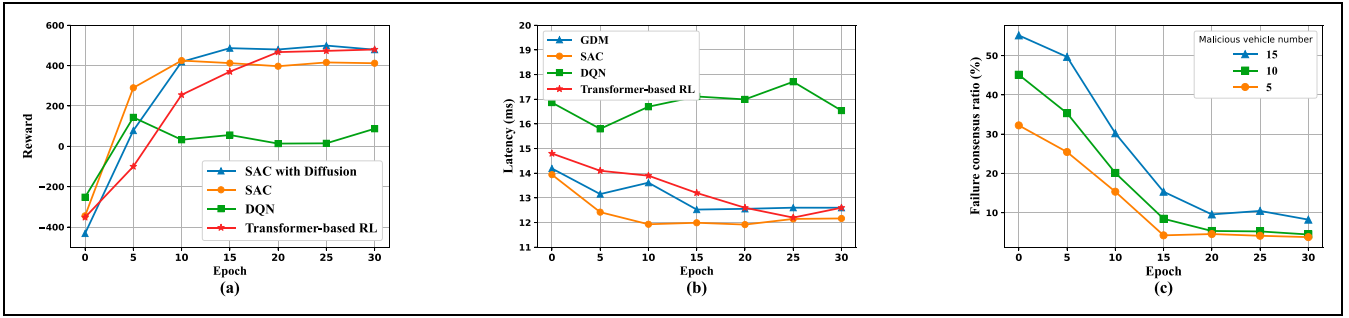


FIG. 3. (a) Environment reward curves. (b) Transaction latency curves. (c) Comparison of vehicle selection frequency as PHs.

communication rates to form a state vector $S = [n, S_t, (c_1, \dots, c_n), (r_1, \dots, r_n)]$, where n is the number of vehicles in the shard, S_t is the average transaction size, c_i is the available computing resource of vehicle i , and r_i is the communication rate between vehicle i and RSU.

2) **Reverse Diffusion for Action Generation:** After obtaining the state vector, it is input into the reverse diffusion process of a GDM as denoising condition information to obtain the decision for blockchain consensus parameter adjustment $a = \{\mathcal{H}^*, S_B^*, I_B^*\}$.

3) **Policy Update:** To train the model, we use the SAC policy to guide model training. The reward for actions is calculated as $R = \alpha \cdot \text{throughput} - \beta \cdot \text{latency}$. If a consensus failure occurs, a negative reward is assigned as a penalty. The policy update method is illustrated in Fig. 2. The noise network within the GDM is implemented using a three-layer neural network, with each layer consisting of 256 neurons. The GDM's denoising step count is 5, and the learning rates for the actor and the critic are set to 1×10^{-4} and 1×10^{-3} , respectively. The weights assigned to throughput and latency are 0.8 and 0.2, respectively.

C. PERFORMANCE EVALUATION

To evaluate the proposed GDM method, we simulate the AVP blockchain environment and implement the GDM model with PyTorch. Specifically, the AVP environment consists of 36 vehicles and 3 shards. The training is set to run for 30 epochs with 1000 transitions per epoch. The average transaction size, the maximum communication rate and the maximum computing resource of vehicles are 10KB, 800Mbps, 2.0G CPU cycles/s, respectively. We used Ethereum transactions from XBlock⁸ to simulate the inter-vehicle transactions and information exchange in the AVP environment.

Fig. 3(a) shows the environment reward for the different optimization algorithms, and Fig. 3(b) shows the average transaction latency per epoch. The results show that GDM outperforms SAC in reward, despite a minor increase in latency. Compared to Transformer-based reinforcement learning [15], GDM has also demonstrated superior convergence performance in discrete action spaces. These suggest that GDM can effectively explore and learn the relationships between state parameters in a complex optimization problem, enabling it to make more effective decisions to balance the trade-off between latency and throughput. However, in the dynamic environment of AVP, the SAC may get trapped in a suboptimal policy.

Fig. 3(c) illustrates the blockchain consensus failure rate under varying numbers of MVs. The GDM remains effective in identifying MVs within the AVP

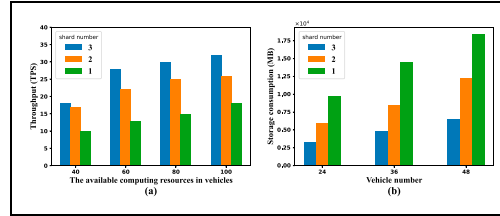


FIG. 4. (a) Throughput under different vehicle computing resource. (b) Storage consumption with different number of shards and vehicles.

system, even when the number of MVs is relatively high. Consequently, this enables an effective reduction in the blockchain consensus failure rate through judicious PH selection.

Furthermore, we evaluate the performance of the sharding blockchain-based AVP framework. Fig. 4(a) compares blockchain throughput with different vehicle computing resources. Due to parallelized intra-shard consensus, the sharding blockchain framework achieves higher throughput than non-sharded, demonstrating sharding's effectiveness for performance improvement in resource-limited scenarios. Fig. 4(b) illustrates the average storage resource consumption of vehicle nodes in AVP. Due to the shard structure, vehicle nodes only need to store the blockchain data of their respective shard, which significantly reduces storage resource consumption compared to a single-shard blockchain. This is particularly important in AVPs, where nodes have limited resources.

VI. FUTURE DIRECTIONS

A. ENHANCING GENAI SECURITY THROUGH BLOCKCHAIN

While GenAI aids blockchain scalability and security, blockchain can also be used to enhance GenAI security, especially in model deployment and inference. It enables decentralized AI model marketplaces for secure sharing, with smart contracts enforcing usage policies for ethical/legal compliance. Additionally, blockchain facilitates decentralized consensus to validate GenAI outputs, reducing adversarial attack risks and ensuring content reliability.

B. EMPOWERING BLOCKCHAIN ECOSYSTEMS WITH GENAI

In addition to enhancing scalability and security, GenAI can improve the functionality and efficiency of the broader blockchain ecosystem. This includes areas like customized smart contract generation and decentralized identity management. For example, in decentralized identity management, GenAI can be used to create and manage self-sovereign identities (SSIs) that are secure, privacy-preserving, and

⁸ <https://xblock.pro/#/>

interoperable across different blockchain platforms, ensuring identity accuracy and enhancing trust and security in decentralized applications, such as Web3, that rely on user identities.

C. SEMANTIC COMMUNICATION AIDED BLOCKCHAIN

As the blockchain ecosystem grows, reducing communication overhead is critical. Semantic communication offers a solution by enabling precise and resource-efficient information transmission. GenAI can leverage this by analyzing and compressing blockchain transaction data, extracting and sending only essential semantic information. For instance, instead of full transaction histories, GenAI can create concise summaries of intents and outcomes, cutting data size while preserving meaning. This is especially useful in high-frequency environments like DeFi or supply chain management.

VII. CONCLUSION

In this paper, we propose the BGI framework, which integrates GenAI with blockchain optimization. Within BGI, we categorize blockchain optimization into three levels: off-chain network, blockchain core, and on-chain application. For each of these three levels, we elaborate on the role of GenAI in optimizing two key performance dimensions: scalability and security. Building on this, we explore the application of GenAI in mainstream blockchain scaling strategies to address the challenges faced by existing approaches. As a case study, we utilize GDMs to optimize the consensus of a sharding blockchain-enabled AVP, dynamically adjusting multiple consensus parameters to enhance the AVP blockchain's throughput and security performance. Finally, we discuss potential future directions for integrating GenAI and blockchain.

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